Flexible demand

The worksheet has three parts, one after each block of the lecture. Each takes about 15 minutes: 10 minutes of work in pairs and 5 minutes in plenary. Hints for the "If you finish early" questions are in the appendix.

Part 1 · The planner 15 min

Find the first-order conditions and say what they mean

The planner chooses thermal capacity K_f, thermal output in the windy state q_f^w, the cost of the last wind site built $\tilde{r}_i$, and consumption q_r^w, $q_r^{\overline{w}}$ (reactive consumers) and $q_{\overline{r}}$ (non-reactive consumers). The wind state w has probability ν, a share β of consumers are reactive. The Lagrangian is

$$\begin{aligned}\mathcal{L} = {} & \underbrace{\beta\left[\nu S(q_r^w) + (1-\nu)S(q_r^{\overline{w}})\right] + (1-\beta)S(q_{\overline{r}})}_{\text{gross surplus}} \\ & \underbrace{-\left(\nu(c+\delta)q_f^w + (1-\nu)(c+\delta)K_f\right)}_{\text{fuel and carbon}} \underbrace{-\left(\overline{K}\int_{\underline{r}_i}^{\tilde{r}_i} r_i\, dF(r_i) + r_f K_f\right)}_{\text{capacity costs}} \\ & + \underbrace{\nu\gamma_w\left[\overline{K}F(\tilde{r}_i) + q_f^w - \beta q_r^w - (1-\beta)q_{\overline{r}}\right]}_{\text{market clearing, } h=w} + \underbrace{(1-\nu)\gamma_{\overline{w}}\left[K_f - \beta q_r^{\overline{w}} - (1-\beta)q_{\overline{r}}\right]}_{\text{market clearing, } h=\overline{w}} \\ & + \underbrace{\nu\left[\underline{\mu}_f q_f^w + \overline{\mu}_f\left(K_f - q_f^w\right) + \underline{\mu}_i\left(\tilde{r}_i - \underline{r}_i\right)\right]}_{\text{bounds}}\end{aligned}$$

γ_w and $\gamma_{\overline{w}}$ are the shadow prices on market clearing, and at the optimum the price in state h is $p^h = \gamma_h$. The μ's are the shadow prices on the bounds $q_f^w \geq 0$, $q_f^w \leq K_f$ and $\tilde{r}_i \geq \underline{r}_i$. *Notation:* $\overline{w}$ is the state without wind. Thermal output has fuel cost c and emission damage δ per unit, and thermal capacity costs r_f per unit. Wind sites have capacity costs $r_i \in [\underline{r}_i, \infty)$ with cdf F and pdf f, and a total potential of $\overline{K}$. All sites with $r_i \leq \tilde{r}_i$ are built.

The class splits in two. Group C (consumers) works on the three consumption choices, and Group P (producers) on thermal output and capacity. In the plenary, each group explains its conditions to the other. Do question 1 before question 2.

Group C: consumers

1. *The conditions.* Differentiate $\mathcal{L}$ and simplify

$$\frac{\partial \mathcal{L}}{\partial q_r^w} = 0, \qquad \frac{\partial \mathcal{L}}{\partial q_r^{\overline{w}}} = 0, \qquad \frac{\partial \mathcal{L}}{\partial q_{\overline{r}}} = 0.$$

2. *The intuition.* What does each condition say? Explain in words. *At most 60 words.*

Group P: producers

1. *The conditions.* Differentiate $\mathcal{L}$ and simplify

$$\frac{\partial \mathcal{L}}{\partial q_f^w} = 0, \qquad \frac{\partial \mathcal{L}}{\partial K_f} = 0.$$

2. *The intuition.* What does each condition say? Which state pays for a MW of thermal capacity? *At most 60 words.*

Stop point

Group C: three conditions and what each says. Group P: two conditions and what each says.

If you finish early

(i) Differentiate $\mathcal{L}$ with respect to the wind threshold and show that

$$\frac{\partial \mathcal{L}}{\partial \tilde{r}_i} = \overline{K} f(\tilde{r}_i)\,(v\gamma_w - \tilde{r}_i) + v\underline{\mu}_i = 0.$$

(ii) What does $\tilde{r}_i = v\gamma_w$ say about the last wind turbine built? When is $\underline{\mu}_i > 0$?

Hint A in the appendix.

Part 2 · A two-state example 15 min

Why exposing consumers to prices still pays

The welfare effect of one more reactive consumer is, from the lecture,

$$\frac{d\mathbb{E}[W]}{d\beta} = \left[\nu S(q_r^{w}) + (1-\nu)S(q_r^{\overline{w}})\right] - S(q_{\bar{r}}) - \nu p^{w}\left(q_r^{w} - q_{\bar{r}}\right) - (1-\nu)p^{\overline{w}}\left(q_r^{\overline{w}} - q_{\bar{r}}\right). \tag{11}$$

Take one consumer with gross surplus $S(q) = 200q - 5q^2$, so demand is $D(p) = 20 - 0.1p$. The two states are equally likely ($\nu = \frac{1}{2}$), with $p^{w} = 40$ and $p^{\overline{w}} = 120$. The fixed retail price is the fair price $p = \nu p^{w} + (1-\nu)p^{\overline{w}} = 80$.

1. *Lemma 2.* Find q_r^{w}, $q_r^{\overline{w}}$ and $q_{\bar{r}}$. Compare the expected gross surplus of a reactive consumer with that of a non-reactive one. Who has more?

2. *Proposition 2.* Compute the two price terms on the second line of (11). Why is their sum positive? Then compute $d\mathbb{E}[W]/d\beta$ and check that it equals $\frac{1}{2}|D'(p)|\,\mathrm{Var}(p^{h})$, where

$$\mathrm{Var}(p^{h}) = \nu(1-\nu)(p^{w} - p^{\overline{w}})^2.$$

   (This formula is our own approximation, not in the paper. It is exact with linear demand.)

   Hint: $|D'(p)| = 0.1$.

3. *Proposition 3.* Suppose more reactive consumers narrow the prices to 60 and 100 while the retail price stays at 80. One consumer is too small to move prices, so take them as given. What is the next reactive consumer worth now, and why?

Stop point

Three quantities, two expected gross surpluses and their difference, the sum of the two price terms in (11), and $d\mathbb{E}[W]/d\beta$ at both price pairs.

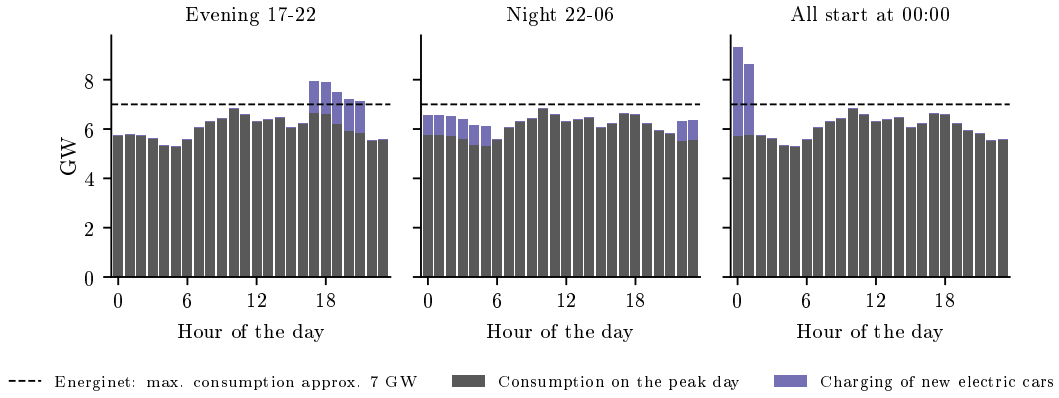
If you finish early

Fabra et al. (2021, Table 1) report a mean price of 10.82 c/kWh with a standard deviation of 1.73, and an average consumption of 0.24 kWh per hour. Use $\frac{1}{2}|D'(p)|\,\mathrm{Var}(p^{h})$ with $|D'(p)| = |\varepsilon|\,q/p$ and $\varepsilon = -0.1$ to compute the gain per household per hour, and then per year (8,760 hours). (Applying the two-state formula across all hours is our own extension, not in the paper. The standard deviation is over all hours in the sample, and $\varepsilon = -0.1$ is larger than the paper's own estimates, so treat the result as an upper bound.) What does the number say about why Spanish households did not respond to the hourly price?

Part 3 · The grid 15 min

What is grid-friendly?

The figure takes the peak consumption day of 2026 and adds charging for the roughly 815,000 electric cars that join the stock by 2030. Without the cars, the peak hour is 6.82 GW. If the cars charge when people come home, the peak is 7.93 GW in the evening. If every car that is plugged in starts at 00:00, when the low tariff begins, the peak is 9.33 GW at night.



Source: own calculations. Peak consumption day (excl. self-consumption and grid losses) plus charging for the roughly 815,000 electric cars added to the stock by 2030 (KF26). Assumptions: 40% charge on a given night with 11 kW. "All start at 00:00" is an upper bound, not a forecast.

1. Propose a rule that defines grid-friendly charging of an electric car.
2. Apply your rule to the figure. What happens if all cars follow it?
3. Which price signal would make the rule unnecessary? Why is it hard to introduce in Denmark?

Stop point

A fixed rule moves the peak. A price that reflects local grid scarcity spreads it.

If you finish early

In the lecture's extension of the model (not in the paper), the grid can carry at most G, and $\lambda_h \geq 0$ is its shadow price. The efficient conditions are $S'(q_r^h) = \gamma_h + \lambda_h$ and $S'(q_{\bar{r}}) = \nu(\gamma_w + \lambda_w) + (1-\nu)(\gamma_{\bar{w}} + \lambda_{\bar{w}})$. Suppose the grid is full only in the windy state: $\lambda_w > 0$ and $\lambda_{\bar{w}} = 0$. The grid company cannot price by state and adds the same tariff t to every hour.

(i) Which t gives non-reactive consumers the efficient condition?

(ii) With that t, what price does a reactive consumer see in each state, compared with $\gamma_h + \lambda_h$? In which state do reactive consumers use too much?

Hint B in the appendix.

Appendix · Hints for "If you finish early"

Hint A (Part 1, wind capacity).

1. $\tilde{r}_i$ appears in three places: the integral in the capacity costs, $F(\tilde{r}_i)$ in market clearing in $h = w$, and the bound $\underline{\mu}_i(\tilde{r}_i - \underline{r}_i)$.
2. Leibniz' rule: $\dfrac{d}{d\tilde{r}_i}\displaystyle\int_{\underline{r}_i}^{\tilde{r}_i} r_i \, dF(r_i) = \dfrac{d}{d\tilde{r}_i}\int_{\underline{r}_i}^{\tilde{r}_i} r_i f(r_i)\, dr_i = \tilde{r}_i f(\tilde{r}_i)$.
3. The derivative of $F(\tilde{r}_i)$ is the density $f(\tilde{r}_i)$. Collect the terms with $\overline{K} f(\tilde{r}_i)$.
4. $\underline{\mu}_i \geq 0$ is the shadow price on the bound $\tilde{r}_i \geq \underline{r}_i$. Write down its Kuhn–Tucker conditions.
5. Interpretation: $\overline{K} f(\tilde{r}_i)$ is the wind capacity added when the cost threshold rises by one unit. What does each unit earn in expectation, and what does it cost?

Hint B (Part 3, the grid).

1. Insert $\lambda_{\overline{w}} = 0$ in the efficient condition for $q_{\overline{r}}$.
2. With the tariff, a non-reactive consumer pays $v\gamma_w + (1-v)\gamma_{\overline{w}} + t$. Set this equal to the right-hand side from step 1 and solve for t.
3. A reactive consumer sees $\gamma_h + t$. Compare with $\gamma_h + \lambda_h$ in w and in $\overline{w}$. Too high a price means too little consumption.
4. A Danish household on a spot-price contract sees the spot price (γ_h) plus a grid tariff fixed in advance by time of day. Is that closer to t or to λ_h?